



EVERYDAY IS A PAYDAY

DATABASE DOCUMENTATION

PayDay — MTN Mobile Money, Orange Money and the App Wallet (Version 1)

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1. Overview

PayDay's data model is intentionally normalized around five core entities: User, Wallet, Linked External Account, Transaction, and Notification. The guiding design principle is that the wallet balance is never edited directly — every change to a Wallet.balance must originate from a Transaction record. This makes the ledger fully auditable: at any point, a wallet's balance can be independently recomputed and verified from the sum of its successful transactions.

Version 1 links each user's wallet to two possible external channels — MTN Mobile Money and Orange Money. The schema is designed so that adding a further channel (such as UBA in a later phase) requires only a new value in the provider enumeration and a corresponding channel adapter, not a change to the core tables.

2. Entity-Relationship Diagram

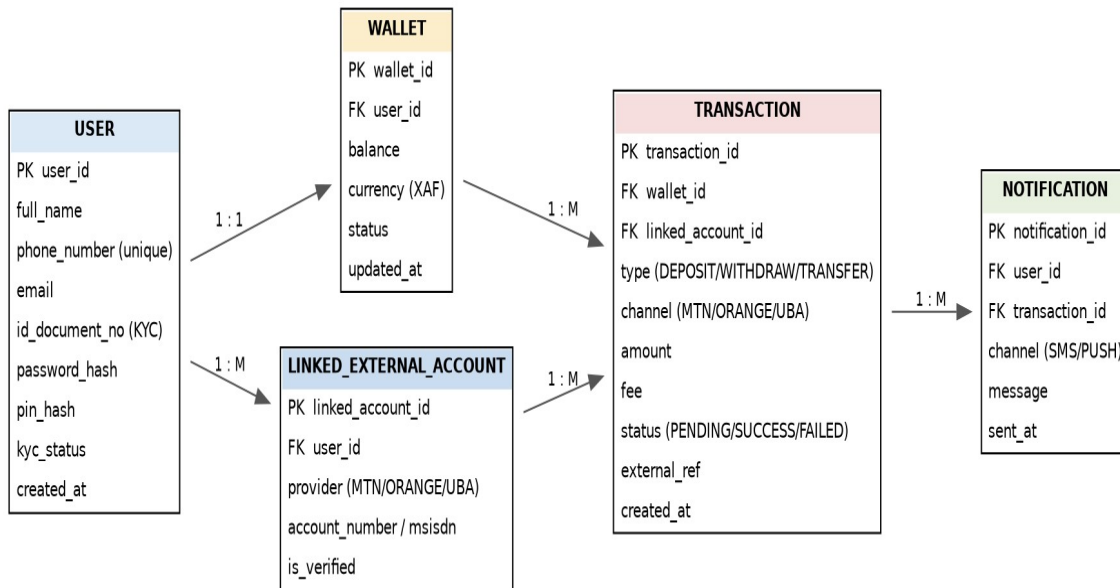


Figure 1: Entity-Relationship Diagram — PayDay

3. Entity Relationships

Relationship	Cardinality	Description
User → Wallet	1 : 1	Each user has exactly one central wallet, created automatically at registration.
User → Linked_External_Account	1 : N	A user may link both an MTN and an Orange account (up to one per provider recommended).
Wallet → Transaction	1 : N	A wallet has many transactions over time; the balance is always the sum of SUCCESS transactions.

Relationship	Cardinality	Description
Linked_External_Account → Transaction	1 : N	Each transaction references exactly one external account as its counterpart channel.
User → Notification	1 : N	A user receives many notifications over time.
Transaction → Notification	1 : N	A transaction may trigger one or more notifications (e.g., SMS and push for the same event).

4. Table Definitions

4.1 User

Stores each registered user's profile, credentials, and KYC status.

Field	Type	Constraints	Description
user_id	UUID / BIGINT	PK	Unique identifier for the user.
full_name	VARCHAR(120)	NOT NULL	User's full legal name, as captured at registration.
phone_number	VARCHAR(20)	UNIQUE, NOT NULL	Primary identifier for login and channel matching (MSISDN).
email	VARCHAR(120)	UNIQUE, NULLABLE	Optional email for receipts and account recovery.
password_hash	VARCHAR(255)	NOT NULL	Bcrypt/argon2 hash — never stored in plaintext.
pin_hash	VARCHAR(255)	NULLABLE	Hashed transaction PIN, set after registration.
id_document_no	VARCHAR(50)	NOT NULL	National ID or passport number, used for KYC.
kyc_status	ENUM	NOT NULL, DEFAULT 'PENDING'	PENDING / VERIFIED / REJECTED.
status	ENUM	NOT NULL, DEFAULT 'ACTIVE'	ACTIVE / SUSPENDED / CLOSED.
created_at	TIMESTAMP	NOT NULL	Account creation time.
updated_at	TIMESTAMP	NOT NULL	Last profile update time.

4.2 Wallet

The central wallet record. Exactly one wallet exists per user, and it is the single source of truth for the user's balance.

Field	Type	Constraints	Description
wallet_id	UUID / BIGINT	PK	Unique identifier for the wallet.
user_id	UUID / BIGINT	FK → User.user_id, UNIQUE	One wallet per user (1:1 relationship).
balance	DECIMAL(14,2)	NOT NULL, DEFAULT 0.00	Current wallet balance. Never edited directly — only via a Transaction.
currency	CHAR(3)	NOT NULL, DEFAULT 'XAF'	ISO currency code; single-currency in Version 1.
status	ENUM	NOT NULL, DEFAULT 'ACTIVE'	ACTIVE / FROZEN / CLOSED.
daily_limit	DECIMAL(14,2)	NOT NULL	Admin-configurable daily transaction ceiling.
monthly_limit	DECIMAL(14,2)	NOT NULL	Admin-configurable monthly transaction ceiling.
created_at	TIMESTAMP	NOT NULL	Wallet creation time (auto-created at registration).
updated_at	TIMESTAMP	NOT NULL	Last balance/status update time.

4.3 Linked External Account

Represents a user's MTN or Orange account, linked to their PayDay wallet so it can be used as a source or destination for transactions.

Field	Type	Constraints	Description
linked_account_id	UUID / BIGINT	PK	Unique identifier for the linked external account.
user_id	UUID / BIGINT	FK → User.user_id	Owner of this linked account.
provider	ENUM	NOT NULL	MTN / ORANGE (UBA reserved for a future version).

Field	Type	Constraints	Description
account_identifier	VARCHAR(30)	NOT NULL	MSISDN or account number on the external provider's system.
is_verified	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether ownership of the external account has been confirmed.
is_default	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether this is the user's default channel for that provider.
created_at	TIMESTAMP	NOT NULL	Date the account was linked.

4.4 Transaction

The heart of the data model — every deposit or withdrawal is recorded here, and this table is the sole authority for wallet balance changes.

Field	Type	Constraints	Description
transaction_id	UUID / BIGINT	PK	Unique identifier for the transaction.
wallet_id	UUID / BIGINT	FK → Wallet.wallet_id	Wallet affected by this transaction.
linked_account_id	UUID / BIGINT	FK → Linked_External_Account.l inked_account_id	External account used as the source or destination.
type	ENUM	NOT NULL	DEPOSIT / WITHDRAW.
channel	ENUM	NOT NULL	MTN / ORANGE.
amount	DECIMAL(14,2)	NOT NULL	Transaction amount, excluding fee.
fee	DECIMAL(14,2)	NOT NULL, DEFAULT 0.00	Fee charged by PayDay for this transaction.
status	ENUM	NOT NULL, DEFAULT 'PENDING'	PENDING / PROCESSING / SUCCESS / FAILED / REVERSED.
external_ref	VARCHAR(100)	NULLABLE	Reference/ID returned by the MTN or Orange API for reconciliation.
failure_reason	VARCHAR(255)	NULLABLE	Populated when status = FAILED.

Field	Type	Constraints	Description
created_at	TIMESTAMP	NOT NULL	Time the transaction was initiated.
completed_at	TIMESTAMP	NULLABLE	Time the transaction reached a final state (SUCCESS/FAILED/REVERSED).

4.5 Notification

Records every SMS or push notification sent to a user, primarily transaction confirmations and alerts.

Field	Type	Constraints	Description
notification_id	UUID / BIGINT	PK	Unique identifier for the notification.
user_id	UUID / BIGINT	FK → User.user_id	Recipient of the notification.
transaction_id	UUID / BIGINT	FK → Transaction.transaction_id, NULLABLE	Related transaction, where applicable.
channel	ENUM	NOT NULL	SMS / PUSH.
message	VARCHAR(255)	NOT NULL	Notification content sent to the user.
status	ENUM	NOT NULL, DEFAULT 'PENDING'	PENDING / SENT / FAILED.
sent_at	TIMESTAMP	NULLABLE	Time the notification was dispatched.

5. Data Integrity Rules

- The Wallet.balance column is never updated directly by application code outside the Wallet Engine — every change must originate from a Transaction row moving to status = SUCCESS.
- A withdrawal places a hold on funds at status = PENDING/PROCESSING; if the external channel fails or times out, the system automatically creates a reversal and releases the hold — history is never edited or deleted.
- external_ref is required before a transaction can move to SUCCESS, so every wallet-affecting event is traceable back to MTN's or Orange's own transaction record for reconciliation.

- Foreign keys enforce that a transaction cannot exist without a valid wallet and a valid linked external account, preventing orphaned financial records.
- kyc_status must be VERIFIED before a user's wallet can send or receive any transaction above a minimal threshold (admin-configurable).

6. Notes

This schema reflects Version 1 scope only (MTN and Orange). The Linked_External_Account.provider enumeration and the Transaction.channel enumeration are the two places a future UBA integration would extend, without requiring structural changes elsewhere in the schema.